

Week 02 Homework -- Banking KPI Design

Zero to Data Engineer Specialization

For each question, follow the five-step framework: understand the KPI, identify the required data, design the model, create the tables, and write the business query. Use the **bank_oltp** dataset you loaded in the Week 02 setup.

Question 1 -- One Big Table (OBT)

Business Requirement

The Head of Retail Banking wants to identify the customers with the highest total account balances.

Step 1 -- Understand the KPI

KPI

Total Customer Balance

Formula

```
SUM(balance) GROUP BY customer
```

Questions to Ask

- What is being measured?
- At what level is it measured?
- Who will consume the report?

Step 2 -- Identify Required Data

What data is needed to calculate the KPI?

Customer

- `customer_id`
- `customer_name`

- city

Account

- account_id
- customer_id
- balance

Step 3 -- Design the OBT

Grain

One row per account.

Table Design

customer_balance_obt
account_id
customer_id
customer_name
city
balance

Step 4 -- Create Tables

> *Students should write the CREATE TABLE statement for customer_balance_obt using the fields in the design above.*

```
CREATE TABLE customer_balance_obt (
    -- your columns here
);
```

Step 5 -- Write the Business Query

```
SELECT
    customer_name,
    SUM(balance) AS total_balance
FROM customer_balance_obt
GROUP BY customer_name
ORDER BY total_balance DESC;
```

Question 2 -- Kimball (Star Schema)

Business Requirement

Management wants to analyze total deposits by branch.

Step 1 -- Understand the KPI

KPI

Total Deposits per Branch

Formula

```
SUM(balance) GROUP BY branch
```

Step 2 -- Identify Required Data

Customer

- customer_id
- customer_name

Branch

- branch_id
- branch_name

Account

- account_id
- balance

Step 3 -- Design the Star Schema

Grain

One row per account balance.

Fact Table

fact_account_balance
account_id (FK)
customer_id (FK)
branch_id (FK)
balance

Dimension Tables

dim_customer
customer_id (PK)
customer_name

dim_branch
branch_id (PK)
branch_name

Step 4 -- Create Tables

> *Students should write CREATE TABLE statements for dim_customer, dim_branch, and fact_account_balance.*

```
CREATE TABLE dim_customer ( ... );
CREATE TABLE dim_branch   ( ... );
CREATE TABLE fact_account_balance ( ... );
```

Step 5 -- Write the Business Query

```
SELECT
    b.branch_name,
    SUM(f.balance) AS total_deposits
FROM fact_account_balance f
JOIN dim_branch b
    ON f.branch_id = b.branch_id
GROUP BY b.branch_name;
```

Question 3 -- Inmon (3NF Enterprise DW)

Business Requirement

Management wants a Customer 360 view that includes both deposits and loans.

Step 1 -- Understand the KPI

KPI

Customer Value

Formula

```
Deposits + Loans
```

Step 2 -- Identify Required Data

- `core_banking.customer`
- `core_banking.account`
- `loans.loan`
- `core_banking.branch`

Step 3 -- Design the 3NF Model

Identify Entities

- `Customer`
- `Account`
- `Loan`
- `Branch`

Identify Relationships

- `Customer → Account`
- `Customer → Loan`
- `Branch  → Account`

> *Normalize to 3NF: each non-key attribute must depend only on the primary key, not on other non-key attributes.*

Step 4 -- Create Tables

> *Students should create all four tables with correct primary keys, foreign keys, and constraints.*

```
CREATE TABLE dim_customer          ( ... );
CREATE TABLE dim_branch            ( ... );
CREATE TABLE fact_account_balance  ( ... );
CREATE TABLE fact_loan             ( ... );
```

Step 5 -- Write the Business Query

```
SELECT
    c.customer_name,
    SUM(ab.balance)
    +
```

```
    COALESCE(SUM(l.loan_amount), 0) AS customer_value
FROM dim_customer c
LEFT JOIN fact_account_balance ab
    ON c.customer_id = ab.customer_id
LEFT JOIN fact_loan l
    ON c.customer_id = l.customer_id
GROUP BY c.customer_name;
```

Question 4 -- Data Vault

Business Requirement

The bank receives customer, account, and loan data from multiple source systems. Management wants a scalable architecture that supports future source systems.

Step 1 -- Understand the KPI

KPI

Customer Exposure

Formula

```
Deposits + Loans
```

Step 2 -- Identify Business Keys

- `customer_id`
- `account_id`
- `loan_id`

Step 3 -- Design the Data Vault

Hubs	Links	Satellites
`hub_customer`	`link_customer_account`	`sat_customer`
`hub_account`	`link_customer_loan`	`sat_account`
`hub_loan`		`sat_loan`

Step 4 -- Create Tables

> *Students should create all eight Data Vault tables with hash keys, business keys, load timestamps, and record sources.*

```
-- Hubs
CREATE TABLE hub_customer ( ... );
CREATE TABLE hub_account  ( ... );
CREATE TABLE hub_loan     ( ... );

-- Links
CREATE TABLE link_customer_account ( ... );
CREATE TABLE link_customer_loan    ( ... );

-- Satellites
CREATE TABLE sat_customer ( ... );
CREATE TABLE sat_account  ( ... );
CREATE TABLE sat_loan     ( ... );
```

Step 5 -- Write the Business Query

> *Create a business view first, then query it.*

```sql
-- Step 1: Create the business view
CREATE VIEW customer_360_view AS
SELECT ... ;   -- join hubs, links, and satellites here

-- Step 2: Answer the business question
SELECT
    customer_id,
    SUM(balance) + SUM(loan_amount) AS customer_exposure
FROM customer_360_view
GROUP BY customer_id
ORDER BY customer_exposure DESC;
```

Think Like a Data Architect

Every question in this homework follows the same five-step thinking process. Internalize this workflow -- it applies to every real-world data engineering problem you will face.

